ASSIGNMENT

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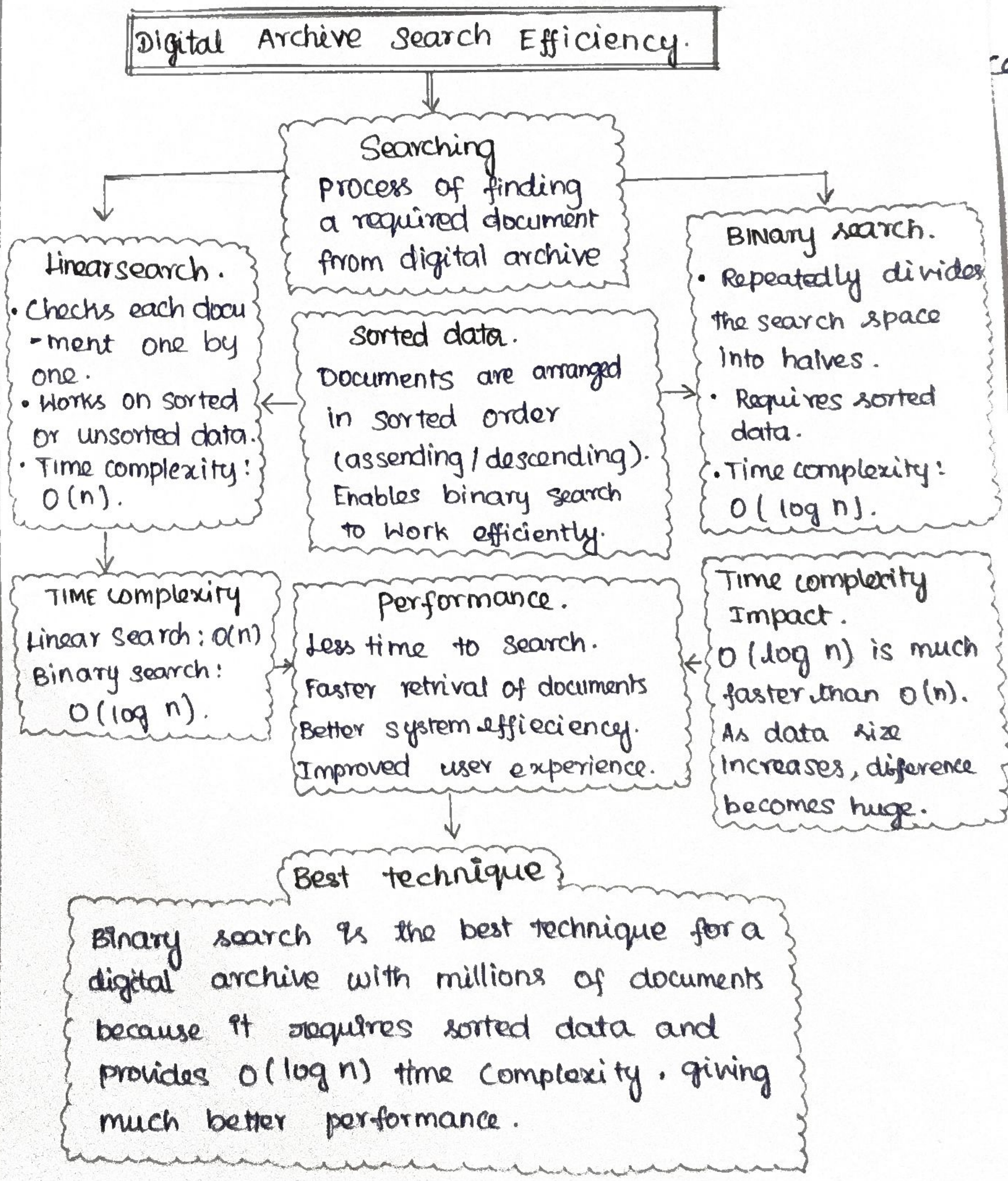
COURSE : DATA STRUCTURE

COURSE CODE : CSA0313

Justification :-

23. A digital archive stores millions of documents. constract a concept map linking searching → linkinger search → Binary search → Sorted data → Time complexity → perfomance. Analyze the impact of data organisation on search efficiency and justify the best technique.



Explanation

* A digital archive containing millions of documents requires an efficiency searching technique. Linear search checks each document one by one, so it works even when the data is unsorted, but it has a time complexity of $O(n)$, making it slow for large datasets.

* Binary search is much faster because it repeatdly divides the search space into two halves. However it can only be used when the documents are arranged in sorted order. Its time complexity is $O(\log n)$, which significantly improves search perfomance.

* The organisation of data has a major impact on search efficiency. Sorted data enables Binary search, reducing search time and improving overall system performance.

Justification :-

For a digital archive with millions of documents, Binary search is the best techique because ;

⇒ It provides faster search with O(log n) time complexity.

⇒ It improves system performance and reduces response time.

⇒ It is ideal for large datasets where the data is maintained in sorted order.

Conclusion :

⇒ organizing data in sorted data in sorted order and using binary search in the most efficient approach for large digital archives.